



Cisco Catalyst SD-WAN CloudOps

First Published: 2019-04-30

Last Modified: 2025-12-16

Americas Headquarters

Cisco Systems, Inc.
170 West Tasman Drive
San Jose, CA 95134-1706
USA
<http://www.cisco.com>
Tel: 408 526-4000
800 553-NETS (6387)
Fax: 408 527-0883



CONTENTS

CHAPTER 1

Cisco CloudOps Overview 1

Types of Fabric Network in Cisco Catalyst SD-WAN 2

Coverage Summary 2

Solution Design 6

Supported Clouds and Cloud Regions 6

Cisco Catalyst SD-WAN Provisioning on Cloud 7

Customer Responsibilities 8

Responsibilities of Cisco CloudOps 9

CHAPTER 2

Ordering, Licensing, and Account Management 11

Role of Cisco Plug and Play 11

Provisioning of Cisco Catalyst SD-WAN Cloud-Hosted Controllers 11

Ordering 12

License types and ordering information 12

A La Carte Ordering 12

EA Ordering 12

Licensing 12

License requirements for various Cisco Catalyst SD-WAN fabric configurations 12

Account Management 14

Transfer Overlay to Another Account 14

On-Premises to Cloud Migration Process Details 15

Cloud-Hosted Controller Deletion Policy 17

Certificate Expiration 18

Abandoned Overlays 18

DNA Subscription Expired 18

Controller Subscription Expired 19

CHAPTER 3

Certificate Management 21

Web Server Certificates 21

Renew Cisco Catalyst SD-WAN SSL Certificates for Controllers 21

CHAPTER 4

Provisioning 25

Getting Access to Cloud Hosted Controllers 25

Cloud-hosted SD-WAN Control Components: Interfaces and access 26

Cloud-hosted SD-WAN Control Components: Interfaces 26

Cloud-hosted SD-WAN Control Components: Access 28

Examples of correct and incorrect configurations for access to SD-WAN Validator 29

Example of correct and incorrect configurations of VPN 0 for access to SD-WAN Validator 29

Custom IP Prefixes for Cloud Hosted Controllers 30

CHAPTER 5

Monitoring 33

Monitor the Cisco Catalyst SD-WAN Cloud-Hosted Controllers 33

Health Monitoring of Overlays with Cisco SD-WAN Manager Version below 20.3.x 34

Health Monitoring of Overlays with Cisco SD-WAN Manager Version Running at or above 20.3.x 34

Alert Notifications by CloudOps 35

Update Overlay Contact for Receiving Alert Notifications 35

CHAPTER 6

Cloud Infrastructure 37

Cisco Cloud-Hosted Controller Snapshots 37

Cisco Catalyst SD-WAN Analytics 38

Pen Test 38

Mandatory Maintenance of Cloud-Hosted Controllers 38

Cisco Catalyst SD-WAN Disaster Recovery Guidelines 39

APPENDIX A

Opening a Cisco TAC Support Case for the Cisco Catalyst SD-WAN Cloud Infra team 43



CHAPTER 1

Cisco CloudOps Overview



Note

To achieve simplification and consistency, the Cisco SD-WAN solution has been rebranded as Cisco Catalyst SD-WAN. In addition, from Cisco IOS XE SD-WAN Release 17.12.1a and Cisco Catalyst SD-WAN Release 20.12.1, the following component changes are applicable: **Cisco vManage to Cisco Catalyst SD-WAN Manager**, **Cisco vAnalytics to Cisco Catalyst SD-WAN Analytics**, **Cisco vBond to Cisco Catalyst SD-WAN Validator**, **Cisco vSmart to Cisco Catalyst SD-WAN Controller**, and **Cisco Controllers to Cisco Catalyst SD-WAN Control Components**. See the latest Release Notes for a comprehensive list of all the component brand name changes. While we transition to the new names, some inconsistencies might be present in the documentation set because of a phased approach to the user interface updates of the software product.

Cisco provides a cloud-hosted subscription service for its Cisco Catalyst SD-WAN Control Components—such as Cisco SD-WAN Manager, Cisco SD-WAN Validator, and Cisco SD-WAN Controller—designed to streamline and speed up deployment. This approach also helps lower the operational costs of managing these components independently. Additionally, the cloud model includes instance monitoring and advanced analytics capabilities.

About This Guide

This document serves as a guide to the Cisco-managed, cloud-hosted Cisco Catalyst SD-WAN Control components, explaining their capabilities and services. Furthermore, it comprehensively details the cloud infrastructure's hosting processes, assigned responsibilities, and pertinent recommendations.

Audience

This document is intended for network design engineers and network operators interested in purchasing or deploying Cisco Catalyst SD-WAN cloud-based subscription options.

- [Types of Fabric Network in Cisco Catalyst SD-WAN](#), on page 2
- [Coverage Summary](#), on page 2
- [Solution Design](#), on page 6
- [Supported Clouds and Cloud Regions](#), on page 6
- [Cisco Catalyst SD-WAN Provisioning on Cloud](#), on page 7
- [Customer Responsibilities](#), on page 8
- [Responsibilities of Cisco CloudOps](#), on page 9

Types of Fabric Network in Cisco Catalyst SD-WAN

- **Cisco SD-WAN Cloud Fabric:** In a Cisco SD-WAN Cloud fabric, Cisco hosts and manages the control components. This fabric type is best for customers who prefer to focus on their edge device networking instead of Cloud control component infrastructure operations.

Some of the salient features of this fabric are:

- Cisco SD-WAN Cloud fabrics always run on the long-lived recommended software releases, providing reliability and stability.

SD-WAN Fabric is mapped to Customer's Smart Account and Virtual account for easier device onboarding, utilizing external management capability of their Virtual Account (VA).

- **Cisco SD-WAN Cloud-Pro Fabric:** In addition to the capabilities of a Cisco SD-WAN Cloud fabric, a Cisco SD-WAN Cloud-Pro fabric allows you to access these options:

- Isolated/Private instance of SD-WAN Control Components.
- Specific software versions.
- AWS or Azure for a Cloud provider and specific location from available Cloud provider regions for deployment of Control Components.
- Ability to choose your Control component software upgrade schedule.
- Commercial Certifications, such as PCI/C5/ENS/CC.
- Government Certifications, such as TxRAMP/StateRAMP.

- **Cisco SD-WAN Cloud-Pro Fabric - Multitenant (for MSPs):** In this type of fabric, the hosting of control components (Cisco SD-WAN Manager, Cisco SD-WAN Validator, and Cisco SD-WAN Controller) is dedicated to the Managed Service Providers. The Managed service provider hosts and manages tenants within this Multitenant environment for their end-customers.



Note A Cisco SD-WAN Cloud-Pro Fabric - Multi-tenant can be hosted only on the AWS cloud provider.

Coverage Summary

Task	Cisco SD-WAN Cloud	Cisco SD-WAN Cloud-Pro	Cisco SD-WAN Cloud-Pro - Multitenant	Comments
Provision Fabric				

Task	Cisco SD-WAN Cloud	Cisco SD-WAN Cloud-Pro	Cisco SD-WAN Cloud-Pro - Multitenant	Comments
Provisioning from Cisco Catalyst SD-WAN Portal	Customer	Customer	Cisco CloudOps	
Monitor and Troubleshoot Cloud Control Components infrastructure				
CPU and data disk utilization	Cisco CloudOps			
Loss of connectivity to network interfaces				
Failure to reach instances				
Monitor Cisco Catalyst SD-WAN services				
Expiry notification of control component SSL certificates	Cisco CloudOps			
Availability of the Cisco SD-WAN Manager web server				
Loss of control connection to the control components				
Capacity management of Cisco Catalyst SD-WAN Controllers	Cisco CloudOps			Cisco CloudOps monitors and upgrades the instance capacity (including expansion to clusters) based on the number of devices on the fabric.
Disaster Recovery				
Take periodic volume-based snapshots	Cisco CloudOps			Note that in multitenancy, the volume-based and config-based snapshot is for the entire multitenancy Cisco SD-WAN Manager cluster, not for any particular tenant.
Take periodic configuration-based backups				
Take On-demand snapshots	Not Applicable	Customer	Customer	
Restore fabric based on volume or configurations	Cisco CloudOps			
Onboard to Cisco SD-WAN Analytics	*Not Applicable	Customer	Customer	Cisco SD-WAN Analytics is onboarded by default for all Cisco Catalyst SD-WAN deployments.

Task	Cisco SD-WAN Cloud	Cisco SD-WAN Cloud-Pro	Cisco SD-WAN Cloud-Pro - Multitenant	Comments
Assist on-premises to cloud migration assistance	Cisco CloudOps			For more details on the On-prem to cloud migration, see On-Premises to Cloud Migration Process Details .
Custom subnets and TACACS	Not Applicable	Customer	Customer	For customers, setting up custom subnets and TACACS is only possible during Day-0 provisioning. For Day-N, customers can open a TAC case with Cisco CloudOps. TACACS is not available for multitenant fabric at present.
Renew control component certificates	Cisco CloudOps	*Customer	*Customer	*CloudOps can help renew certificates on customer request.
Upgrade software				
Control component software upgrade	Cisco CloudOps	*Cisco CloudOps	*Cisco CloudOps	*Cisco performs upgrades only to recommended releases.
Edge device/node software upgrade	Customer			
Upload and manage Edge images in Cisco SD-WAN Manager Software Repository	Cisco CloudOps	Customer	Customer	
Respond to Cisco CloudOps notifications to authorize the service window, instance reboot, review, or verify changes carried out by Cisco CloudOps	Customer			
Create Smart Accounts (SA) or Virtual Accounts (VA) on <code>software.cisco.com</code> and attach Cisco Catalyst SD-WAN subscribed devices to the SA/VA	Customer			
Allow external management of SA/VA on PNP Connect	Customer	Not Applicable	Not Applicable	Do not allow external management of SA/VA on PNP Connect before provisioning a fabric in Cisco Catalyst SD-WAN Portal. The provisioning workflow automatically enables the external management.

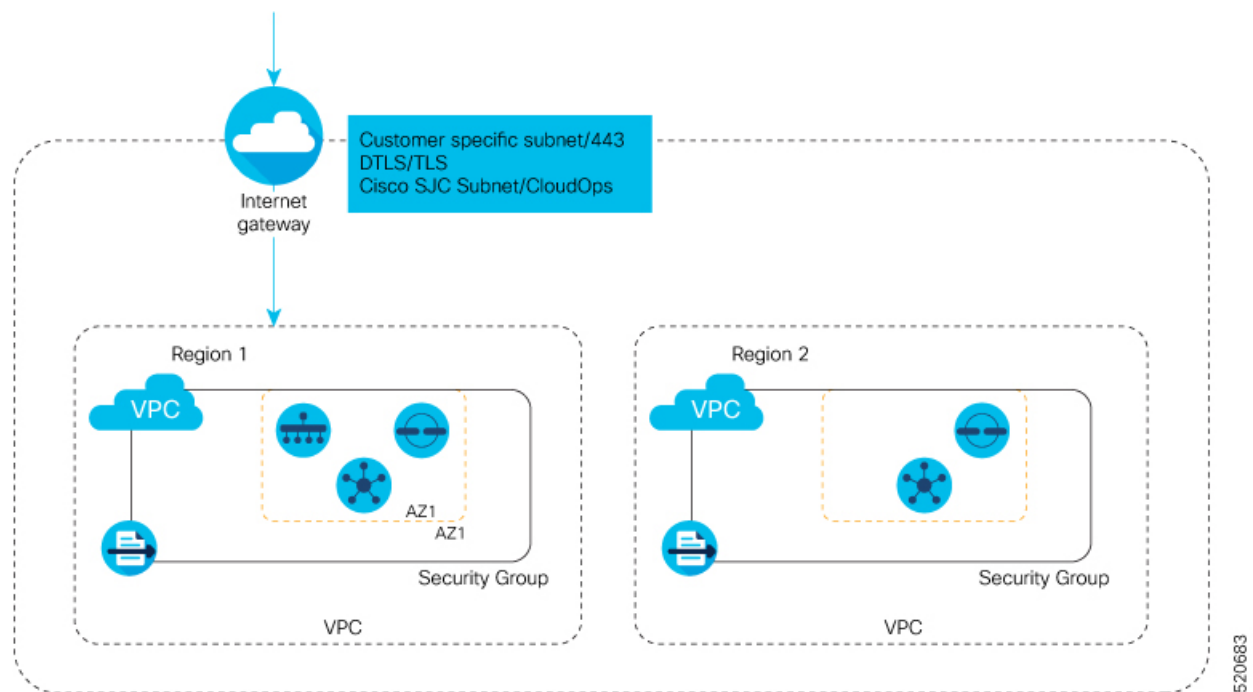
Task	Cisco SD-WAN Cloud	Cisco SD-WAN Cloud-Pro	Cisco SD-WAN Cloud-Pro - Multitenant	Comments
Accept external management of SA/VA and map tenant VA to customer SA/VA	Cisco CloudOps	Not Applicable	Not Applicable	
Define device configuration templates and policies through Cisco SD-WAN Manager	Customer			
Perform other activities that require logging in to Cisco SD-WAN Manager, such as template and policy configuration and edge device management	Customer			
Manage web server certificates	Cisco CloudOps	*Customer	**Customer	*CloudOps can help renew certificates on customer request. **CloudOps can renew web certificates if the MT fabric is deployed in the cisco.com domain.
Sync edge serials with credentials	*Not Applicable	Customer	Customer	*Cisco SD-WAN Cloud customers use their Cisco Connection On-line (CCO) login credentials for Single-Sign-On and syncing edge serials.
Managed Allowed IP access list	Not Applicable	Customer	Customer	
Custom Identity Provider (IdP) Configuration	Not Applicable	Customer	Customer	Cisco SD-WAN Cloud only supports Cisco Connection On-line (CCO) as an identity provider. Customers can use Single-Sign-On to navigate among Catalyst SD-WAN applications such as Cisco SD-WAN Manager, Cisco SD-WAN Analytics, and Cisco Catalyst SD-WAN Portal.

Solution Design

About This Solution

When you choose a cloud-based subscription for your Cisco Catalyst SD-WAN Control Components, Cisco deploys Cisco SD-WAN Manager, Cisco SD-WAN Validator, and Cisco SD-WAN Controller on the public cloud. Cisco then provides you with administrator access. By default, a single Cisco SD-WAN Manager, Cisco SD-WAN Validator, and Cisco SD-WAN Controller are deployed in the primary cloud region and an additional Cisco SD-WAN Validator and Cisco SD-WAN Controller are deployed in the secondary or backup region.

Figure 1: Solution Architecture



Supported Clouds and Cloud Regions

The following clouds and cloud regions are supported for Cisco Catalyst SD-WAN Control Component deployments:

Table 1: Supported clouds and cloud regions for SD-WAN Cloud

Amazon Web Services
APAC
EU
US

Table 2: Supported clouds and cloud regions for Cisco SD-WAN Cloud-Pro

Amazon Web Services	Microsoft Azure
Asia Pacific—Jakarta Indonesia	Asia Pacific Australia East—Sydney New South Wales
Asia Pacific—Mumbai India	Asia Pacific Australia Southeast—Melbourne Victoria
Asia Pacific—Hyderabad India	Asia Pacific Japan East—Tokyo
Asia Pacific—Seoul South Korea	Asia Pacific Southeast Asia—Singapore
Asia Pacific—Singapore Singapore	Asia Pacific West India—Mumbai
Asia Pacific—Sydney Australia	Asia Pacific South India
Asia Pacific—Melbourne Australia	UAE North—Dubai
Asia Pacific—Tokyo Japan	Asia Pacific Australia Central—Canberra
Africa—Cape Town	South Africa—North
Canada Central—Montreal Canada	Canada Central—Montreal Canada
Canada West—Calgary Canada	Canada East
EU—Frankfurt Germany	Americas Brazil South—Sao Paulo State
EU—Ireland Dublin	Europe France Central—Paris
EU—London UK	Europe North Europe—Ireland
EU—Stockholm Sweden	Europe UK South—London
South America—Sao Paulo Brazil	Europe West Europe—Netherlands
US East—Northern Virginia USA	Americas East US—Virginia
US West—Northern California USA	Americas West US—California
US West—Oregon USA	Americas West US 2—Washington

Cisco Catalyst SD-WAN Provisioning on Cloud



Note By default, Cisco provisions one Cisco SD-WAN Manager, two Cisco SD-WAN Validators and two Cisco SD-WAN Controllers for a fabric of size <1500.

For information on recommended computing resources, see [Recommended Computing Resources for Cisco Catalyst SD-WAN Control Components](#).

Customer Responsibilities

- Manage allowed access-list with your source public IP ranges for management access of control components.
- Renew control component certificates on time.
- Before making any changes in the Cisco Catalyst SD-WAN Portal, take the on-demand snapshot using the procedure, [Take an On-Demand Snapshot](#) and configuration backup using [Back Up the Active Cisco SD-WAN Manager](#) procedure.
- Upgrade the software.
 - You can open a TAC case for the following:
 - If you face any issues with software upgrade.
 - If you want any rollback.
 - The Cisco SD-WAN Validator and Cisco SD-WAN Controller are stateless services. Therefore, you do not need to take backups for these services. Cisco SD-WAN Manager automatically pushes the configurations once they are attached to templates.

We recommend that you create and attach templates to the Cisco SD-WAN Validator and Cisco SD-WAN Controller instead, so the Cisco SD-WAN Manager backups automatically include the configuration backup of the control components.
 - The Cisco Catalyst SD-WAN support teams can assist with the control component software upgrade for all deployment types.
 - It is your responsibility to upgrade the software version of an edge device. For the compatible versions of edge devices based on control component versions, see [Cisco SD-WAN Controller Compatibility Matrix](#).
- Respond to the notifications sent by Cisco CloudOps to authorize the service window, instance reboot, review, or verify changes carried out by Cisco CloudOps.
- For a Cisco SD-WAN Cloud-Pro fabric, configure the third interface on Cisco SD-WAN Manager with static IP or DHCP based IP to use it for SD-AVC. By default the third interface is in shut state.
- Open a TAC case to arrange a service window when you receive a notification from Cisco CloudOps. Some operations can be performed only with the consent of the customer.
- Create Smart Accounts (SA) or Virtual Accounts (VA) on `software.cisco.com` and attach Cisco Catalyst SD-WAN subscribed devices to the SA or VA.
- Define device configuration templates and policies through Cisco SD-WAN Manager.
- Perform other activities that require logging in to Cisco SD-WAN Manager.
- For a Cisco SD-WAN Cloud fabric, open a Cisco TAC support case if you need specific software versions to be added in the Cisco SD-WAN Manager software repository.

Your failure to meet the responsibilities outlined in this section will invalidate the [SD-WAN Cloud SLA](#), including any guaranteed service uptimes.

Responsibilities of Cisco CloudOps

Fabric Provisioning

- Provision cloud-hosted control components for your Cisco Catalyst SD-WAN fabric, configure a unique admin password with an expiry time of a week, and hand over Cisco SD-WAN Manager to you.
- Configure Cisco SD-WAN Manager with a default template and policy when you choose the default template and policy push option on the sales order.
- Create and manage Cisco SD-WAN Cloud, Cisco SD-WAN Cloud-Pro, and MT clusters as needed.
- Manage tenants on multitenant fabrics (for direct enterprise customers).

Monitor and Troubleshoot

Cisco CloudOps monitors the health of cloud-hosted fabric and troubleshoots if there are any issues.

- Cisco CloudOps is backed by a real-time monitoring system that checks the health of Cisco Catalyst SD-WAN control components and generates alerts. The check includes the health of Cisco SD-WAN Manager, application or web server, other micro services, and configuration or statistics databases.
- Take proactive action for cloud infrastructure issues, which are beyond your control. Else, notify you about the potential issues and request you to open a Cisco TAC support case for further investigation.
- Manage alerts based on notifications from the cloud provider environments on instance up or down states and CPU or network inactivity status.
- Resolve the alerts proactively if it doesn't require a down time of the services. Notify you when services flap.
- Send 30-, 15-, and 5-day notices to you to renew expiring certificates on Cisco SD-WAN Manager. Cisco Catalyst SD-WAN control component certificates have a validity of one year.

Cloud Infrastructure Support

- Carry out disaster recovery workflows, including snapshot volumes or configurations. Restore Cisco SD-WAN Manager clusters based on volumes or configurations.
- Provision custom subnetting to extend your on-premises network into the cloud-hosted fabric network.
- Manage on-premises to cloud migrations.

Capacity Management

- Monitor the growth of devices per fabric along with the control component instance capacity parameters such as CPU, disk, and memory utilizations. Follow a pre-set guideline to increase the capacity of the service instances as needed.



CHAPTER 2

Ordering, Licensing, and Account Management



Note

To achieve simplification and consistency, the Cisco SD-WAN solution has been rebranded as Cisco Catalyst SD-WAN. In addition, from Cisco IOS XE SD-WAN Release 17.12.1a and Cisco Catalyst SD-WAN Release 20.12.1, the following component changes are applicable: **Cisco vManage to Cisco Catalyst SD-WAN Manager**, **Cisco vAnalytics to Cisco Catalyst SD-WAN Analytics**, **Cisco vBond to Cisco Catalyst SD-WAN Validator**, **Cisco vSmart to Cisco Catalyst SD-WAN Controller**, and **Cisco Controllers to Cisco Catalyst SD-WAN Control Components**. See the latest Release Notes for a comprehensive list of all the component brand name changes. While we transition to the new names, some inconsistencies might be present in the documentation set because of a phased approach to the user interface updates of the software product.

- [Role of Cisco Plug and Play, on page 11](#)
- [Ordering, on page 12](#)
- [Licensing, on page 12](#)
- [Account Management, on page 14](#)

Role of Cisco Plug and Play

Cisco Plug and Play replaces the legacy process of Cisco Catalyst SD-WAN Salesforce (SFDC).

Refer to the following guide for information about Cisco Catalyst SD-WAN Plug and Play:

- [Cisco Plug and Play Support Guide](#)
- [FAQs](#)

Provisioning of Cisco Catalyst SD-WAN Cloud-Hosted Controllers

The Cisco SD-WAN Portal allows creation of the Cisco Catalyst SD-WAN cloud-hosted controllers for a sales order after the following conditions are met:

1. The sales order has the cloud subscription licenses for edge nodes and Cisco SD-WAN Control Components. SKUs are only needed for additional paid controllers.
2. Cisco Catalyst SD-WAN items in the sales order are marked as **Shipped**.

- The sales order is assigned to an active Smart Account (SA), and, within that SA, to a Virtual Account (VA).

Ordering

License types and ordering information

There are two types of licenses and contracts.

- **A La Carte:** Purchase DNA Cloud (DNA-C) licenses and Cisco Catalyst SD-WAN Controller stock keeping units (SKUs) (if required).
- **Enterprise Agreement (EA):** Purchase an EA bundle that includes Cisco Catalyst SD-WAN Controller SKUs (if required).

A La Carte Ordering

For customers who prefer to purchase a la carte licenses for Cisco Catalyst SD-WAN Controller, see [Cisco Catalyst SD-WAN Controller Ordering Guide](#).

EA Ordering

For provisioning a Cisco Catalyst SD-WAN cloud-hosted controller for an Enterprise Agreement (EA) customer, place a request on the EA Workspace (EAWS).

Licensing

License requirements for various Cisco Catalyst SD-WAN fabric configurations

First fabric

Table 3: First fabric associated with the Smart Account

For these requirements...		Here's what you need...		
SD-WAN fabric requirements	Certified environment	SD-WAN Control Components SKU requirements	Device license requirements	Comments
Cisco SD-WAN Cloud	No	None	DNA-C license for each device. See Note (2).	

For these requirements...		Here's what you need...		
SD-WAN fabric requirements	Certified environment	SD-WAN Control Components SKU requirements	Device license requirements	Comments
Cisco SD-WAN Cloud-Pro (See Note (1))	No	Check with your Cisco Sales representative or channel partner. See Note (3).	DNA-C license for each device. See Note (2).	Check with your Cisco Sales representative to determine whether your network specifications require a Cisco SD-WAN Cloud-Pro fabric.
	Yes	Check with your Cisco Sales representative or channel partner. See Note (3).	DNA-C license for each device. See Note (2).	A certified environment requires a Cisco SD-WAN Cloud-Pro fabric.

(1): Setting up a Cisco SD-WAN Cloud-Pro fabric requires opening a [Cisco TAC](#) case with Cisco SD-WAN Cloud Infra Team.

(2): See the [Cisco DNA Software for SD-WAN and Routing Ordering Guide](#) for information about ordering DNA-C licenses.

(3): See the [Cisco Catalyst SD-WAN Control Components Ordering Guide](#).

Additional fabrics

Table 4: Second or subsequent fabrics associated with the Smart Account

For these requirements...		Here's what you need...		
SD-WAN fabric requirements	Certified environment	SD-WAN Control Components SKU requirements	Device license requirements	Comments
Cisco SD-WAN Cloud	No	None	DNA-C license for each device. See Note (2).	

For these requirements...		Here's what you need...		
SD-WAN fabric requirements	Certified environment	SD-WAN Control Components SKU requirements	Device license requirements	Comments
Cisco SD-WAN Cloud-Pro (See Note (1))	No	Check with your Cisco Sales representative or channel partner. See Note (3).	DNA-C license for each device. See Note (2).	Check with your Cisco Sales representative to determine whether your network specifications require a Cisco SD-WAN Cloud-Pro fabric.
	Yes	SD-WAN Control Components license required. See Note (3).	DNA-C license for each device. See Note (2).	A certified environment requires a Cisco SD-WAN Cloud-Pro fabric.

(1): Setting up a dedicated fabric requires opening a [Cisco TAC](#) case with Cisco SD-WAN Cloud Infra Team.

(2): See the [Cisco DNA Software for SD-WAN and Routing Ordering Guide](#) for information about ordering DNA-C licenses.

(3): See the [Cisco Catalyst SD-WAN Control Components Ordering Guide](#).

Account Management

Transfer Overlay to Another Account

To move an overlay from one Smart Account (SA) or Virtual Account (VA) to another SA or VA:

- Open a Cisco TAC support case for the migration request.
- Specify the SA and VA details for both the source and destination in the [Cisco TAC](#) case.

There is no downtime expected for this migration.

You can move the device serial numbers to the new SA or VA using the PNP **Transfer Selected** button, or you can open a Cisco TAC support case for assistance.

The functionality and the following details of the overlay do not change during this migration:

1. Organization name
2. Cisco SD-WAN Validator, Cisco SD-WAN Manager, or Cisco SD-WAN Controller DNS name
3. All current IPs assigned to all controllers
4. The entire Cisco SD-WAN Manager configuration, including certificates

5. Current allowed list of IP addresses

After the overlay migration, you may need to update the SA credentials configured in the Cisco SD-WAN Manager settings.

On-Premises to Cloud Migration Process Details

In the case, where an existing on-premise Cisco Catalyst SD-WAN overlay needs to be migrated to Cisco-provisioned cloud-hosted controllers, the process is outlined below:

**Note**

This migration process is only supported for on-premise single tenant overlays to a cloud-hosted single tenant overlay controller set. This migration is not supported for shared tenant or multi-tenant overlays.

Overall Process

- Purchase Cisco DNA subscriptions for cloud and controller SKUs for cloud.
- You must open a Cisco TAC support case with the Cisco CloudOps team and request for the on-premises to cloud migration.
- You must provide details about the following:
 - Existing Smart Account (SA) and Virtual Account (VA) where the on-premises overlay controller profile is created.
 - The sales order number where cloud subscriptions were purchased.
 - Current on-premises configured organization name of overlay.
 - Choice of the required cloud type.
 - Choice of the required primary and secondary region of provisioning.
 - Single email address as contact for receiving alert notifications and other communications from the Cisco CloudOps team (team email address is preferred).
 - Optional choice of hostname for the FQDN of the Cisco SD-WAN Manager and the Cisco SD-WAN Validator to be provisioned.
 - Optional choice of custom private IP subnets required for TACACS/AAA/Syslog or other such use cases (provide a /24 IP prefix for each of the two regions of provisioning).
 - Current on-premises overlay fabric size in terms of number of edges deployed.
 - Current on-premises overlay Cisco SD-WAN Manager, Cisco SD-WAN Validator, and Cisco SD-WAN Controller instances running software versions.
 - Current on-premises overlay controller certificate source (Cisco/Symantec/Enterprise) root CA.
 - Configuration database backup copy from the current on-premises overlay Cisco SD-WAN Manager.



Note You can either reset the Cisco SD-WAN Manager configuration database password to the default and then take the backup, or take the backup with your configured password and share that password on the Cisco TAC case.

- Copy of the running configuration from the current on-premises overlay Cisco SD-WAN Manager
- Range of system-IP addresses to be used for cloud-hosted controllers (should be an unused range within the current on-premises Cisco Catalyst SD-WAN fabric).
- The Cisco CloudOps team provisions the cloud-hosted controller set, installs controller certificates, and shares details.
- The Cisco CloudOps team applies the configuration database backup and the running configuration provided from the on-premises Cisco SD-WAN Manager to the new cloud-hosted Cisco SD-WAN Manager instance.
- You may need to update your enterprise firewalls as required, with the new IPs of the cloud-hosted controllers.
- Set up and execute a pilot change window to migrate one or more test edge nodes to the cloud-hosted controllers and then roll back to the on-premises Cisco SD-WAN Manager.
- Migration is triggered by configuring the new Cisco SD-WAN Validator FQDN on the edge node.
- Take necessary measures to prepare for the final change window.
- Set up and execute a final change window to migrate all edge nodes from on-premises to cloud-hosted controller set.
- If templates were created and applied for the on-premises Cisco SD-WAN Manager, Cisco SD-WAN Validators, and Cisco SD-WAN Controllers, then they must be reviewed and corrected, before applying them to the cloud-hosted controllers, post migration. Special care must be taken with respect to the interface configuration.
- The edge templates that are created and applied for the on-premise Cisco SD-WAN Manager contain the pre-migration orchestrator FQDN when the database is restored to the new Cisco SD-WAN Manager. You must update the target Cisco SD-WAN Manager to reflect the post-migration orchestrator FQDN.

Prerequisites

- Before opening a case, you must upgrade all your existing controllers and edge nodes to one of the latest Cisco-suggested release versions and verify that your data plane is stable.
- You must have all edge nodes attached to a template or agree to reconfigure the edge nodes manually for the migration.
- You must have all edge nodes with working NTP and DNS.
- You must provide the root CA to Cisco if in case you are using enterprise certificates on the on-premises controllers.
- You must have out-of-band access to edge nodes via console or an alternate way in case the edge nodes need manual configuration for recovery.

Considerations and Impact

- You must work with your Cisco Account Team or Cisco support to procure Cisco Catalyst SD-WAN cloud subscriptions and add them to the existing Smart Account (SA) and Virtual Account (VA) where the on-premises overlay controller profile is created.

- The Cisco CloudOps team provisions Cisco SD-WAN Manager only in the primary region.

There is a Cisco SD-WAN Validator and Cisco SD-WAN Controller instance provisioned in both the primary and the secondary regions.

- The Cisco CloudOps team creates a new controller profile in the same SA/VA as the existing on-premises overlay.

This allows the cloud-hosted controller set to have the same organization name as the existing on-premises overlay. This in turn makes it possible to transfer the configuration database from on-premises Cisco SD-WAN Manager to the cloud-hosted Cisco SD-WAN Manager.

The configuration database restore method, otherwise, can't be used if the source and destination Cisco SD-WAN Manager instances have different organization name configured. Organization name on a cloud-hosted Cisco SD-WAN Manager instance can't be changed once provisioned.

- As the new Cisco SD-WAN Manager is configured using the configuration database restore method, the statistics database from the on-premises Cisco SD-WAN Manager will not be migrated.
- If Cisco SD-WAN Analytics is in use on the on-premises overlay, it continues to work.

There may be some data loss when the migration happens, as the new cloud Cisco SD-WAN Manager starts fresh data collection and sends it to the Cisco SD-WAN Analytics servers.

- As the Cisco SD-WAN Validator FQDN changes, the configuration on the edge nodes requires to be updated for the migration.

This can be done via CLI templates from Cisco SD-WAN Manager applied to all the edge nodes. If no CLI templates exist on the on-premises Cisco SD-WAN Manager, you must create and apply them before starting the migration. If you do not prefer CLI templates, then you would need to manually reconfigure all the edge nodes individually via console or ssh.

- If any issue occurs during the edge node migration, you may need to have an out-of-band management access to the edge nodes to make changes manually to switch over to new Cisco SD-WAN Validators.
- At the time of migration, the control and data plane flaps for each edge node as it is pointed to the new Cisco SD-WAN Validator DNS and reconnects to the new cloud-hosted controllers.
- It is mandatory that all edge nodes be configured with working NTP and DNS before the migration.
- Rollback plan would involve Cisco SD-WAN Validator configuration to be changed back on the edge nodes to the on-premises Cisco SD-WAN Validator.
- After successful migration, the controller profile that you hosted can be deleted from Cisco PNP SA/VA.

Cloud-Hosted Controller Deletion Policy

Cisco can delete a customer cloud-hosted controller overlay based on the following conditions:

Certificate Expiration

- **Identification Stage:** If your controller certificates have been expired for 15 days or more, and you have not renewed the certificates, Cisco can move your cloud-hosted controller to a shutdown state. The expired controller certificates indicate that the cloud-hosted controller overlay and the connected devices are not being used.
- **Final Termination:** If your fabric remains in the shutdown state for a period of at least 30 days, and if you have not made any communication to Cisco to recover the controllers, Cisco deletes the controllers. As a result, your data cannot be recovered.
- **Reprovisioning:** After a fabric is removed, it needs to be reprovisioned. If you have an active Cisco Digital Network Architecture (Cisco DNA) license, you can request a new fabric.

Abandoned Overlays

- **Identification Stage:** If the cloud-hosted controllers are provisioned for six months or more and
 - there are no active edge devices, or
 - the overlays are in the shutdown state for 30 days or more for reasons other than those set forth in this Cloud-Hosted Controller Policy,

then Cisco can deem your cloud-hosted controller as abandoned. Please note that no active edge devices or shutdown overlays indicate that the Cisco Catalyst SD-WAN overlay and the cloud-hosted controller devices are not being used.

- **Notification Stage:** Cisco sends notifications to you communicating the overlay abandoned state along with a target shutdown date.
- **Shutdown Stage:** If the customer overlay continues to remain unused even after the notifications, Cisco shuts down the overlay on the specified date.
- **Final Termination:** If you have not communicated to Cisco to recover Cisco Catalyst SD-WAN cloud-hosted controllers within 30 days of the overlay shutdown, Cisco deletes the controllers. As a result, the customer data cannot be recovered.
- **Reprovisioning:** Once an overlay is deleted, it needs to be reprovisioned. If you have an active Cisco Digital Network Architecture (Cisco DNA) license, you can request a new cloud-hosted controller overlay.

DNA Subscription Expired

This policy applies to Cisco Digital Network Architecture (Cisco DNA) subscriptions for the devices licensed before Cisco made the cloud controller subscription separately available. It is also known as Pre-Controller Subscription Offering.

- **Identification Stage:** If all the Cisco DNA subscriptions for your devices connected to the cloud-hosted controller have expired, Cisco can deem your corresponding cloud-hosted controller as subscription expired.
- **Notification Stage:** Cisco sends notifications to you communicating the overlay abandoned state along with a target shutdown date. Ensure that you keep your contact information up-to-date to receive timely notifications.

- **Shutdown Stage:** If the customer overlay continues to run with the expired DNA subscriptions even after the notifications, Cisco shuts down the overlay on the specified date.
- **Final Termination:** If you have not communicated to Cisco to recover your Cisco Catalyst SD-WAN cloud-hosted controllers within 30 days of the overlay shutdown, Cisco deletes the controllers. As a result, the customer data cannot be recovered.
- **Reprovisioning:** Once an overlay is deleted, it needs to be reprovisioned. You can purchase a new cloud-hosted controller overlay by purchasing the required stock keeping units (SKUs).

Controller Subscription Expired

A controller subscription is licensed separately from the Cisco Digital Network Architecture (Cisco DNA) subscriptions for devices.

- **Identification Stage:** If the subscription of your cloud-hosted controllers has expired, and if you have not renewed it, Cisco can deem your corresponding cloud-hosted controller as subscription expired.
- **Notification Stage:** Cisco sends notifications to you communicating the overlay abandoned state along with a target shutdown date. Ensure that you keep your contact information up-to-date to receive timely notifications.
- **Shutdown Stage:** If the controller subscription continues to remain unrenewed even after the notifications, Cisco shuts down the overlay on the specified date.
- **Final Termination:** If you have not communicated to Cisco to recover your Cisco Catalyst SD-WAN cloud-hosted controllers within 30 days of the overlay shutdown, Cisco deletes the controllers. As a result, the customer data cannot be recovered.
- **Reprovisioning:** Once an overlay is deleted, it needs to be reprovisioned. You can purchase a new cloud-hosted controller overlay by purchasing the required stock keeping units (SKUs).



Note

Failure to renew your DNA subscription for the Cisco cloud-hosted controllers may impact the functionality of the Cisco Catalyst SD-WAN features that are part of the Cisco DNA subscription for your devices. It is because these features are dependent on Cisco SD-WAN Controllers.



CHAPTER 3

Certificate Management



Note To achieve simplification and consistency, the Cisco SD-WAN solution has been rebranded as Cisco Catalyst SD-WAN. In addition, from Cisco IOS XE SD-WAN Release 17.12.1a and Cisco Catalyst SD-WAN Release 20.12.1, the following component changes are applicable: **Cisco vManage** to **Cisco Catalyst SD-WAN Manager**, **Cisco vAnalytics** to **Cisco Catalyst SD-WAN Analytics**, **Cisco vBond** to **Cisco Catalyst SD-WAN Validator**, **Cisco vSmart** to **Cisco Catalyst SD-WAN Controller**, and **Cisco Controllers** to **Cisco Catalyst SD-WAN Control Components**. See the latest Release Notes for a comprehensive list of all the component brand name changes. While we transition to the new names, some inconsistencies might be present in the documentation set because of a phased approach to the user interface updates of the software product.

- [Web Server Certificates](#), on page 21
- [Renew Cisco Catalyst SD-WAN SSL Certificates for Controllers](#), on page 21

Web Server Certificates

Cisco does not issue web certificates for Cisco SD-WAN Manager. We recommend that you generate the Certificate Signing Request (CSR) and get it signed by your Certificate Authority (CA) for your Domain Name System (DNS) name. Then, you may either add an A entry in your DNS server for the IP, or a CNAME to the `.viptela.net` / `.sdwan.cisco.com` Cisco SD-WAN Manager DNS name for commercial deployments or `.sdwangov.fedramp.cisco` for government deployments.



Note The controller certificates issued by Cisco are for the controllers to use internally. You cannot use these certificates to issue web server certificates.

For more information, see the [Web Server Certificates](#) section in the Cisco Catalyst SD-WAN Getting Started Guide.

Renew Cisco Catalyst SD-WAN SSL Certificates for Controllers

Signed certificates are used to authenticate devices in the overlay network. After being authenticated, devices can establish secure sessions between each other.



Note The certificate renewal process is applicable only if you have a Cisco SD-WAN Cloud-Pro single tenant or multi-tenant controller overlay. This process is not applicable if you have a shared tenant overlay.

You can generate the Certificate Signing Request (CSR) as well as install the signed certificates, using Cisco SD-WAN Manager. There are 3 options for Certificate Root CA:

1. Cisco Root CA bundle (already present on controllers with software version 19.2.3 and above, Cisco Catalyst SD-WAN devices with software version 19.2.3 and above, Cisco IOS XE Catalyst SD-WAN devices with software versions 16.12.3+ or 16.10.4+ or 17.x+).
2. Symantec/Digicert Root CA (already present on all controllers, Cisco Catalyst SD-WAN devices and Cisco IOS XE Catalyst SD-WAN devices).
3. Your own Enterprise Root CA.



Note Select the certificate-generation method only once. The method you select is automatically applied each time you add a device to the overlay network.

To renew the controller certificates, you need to follow the appropriate process based on your deployment type and certificate type:

- The controller certification authorization settings configure the certification- generation process for all controller devices. For more information, see [Cisco Catalyst SD-WAN Controller Certificates](#).
- Note that since the certificate renewal involves an entire control plane flap, you are required to follow the instructions as per above, to renew the certificates, even for Cisco SD-WAN Cloud-Pro controllers.
- The Cisco CloudOps team does not automatically renew the certificates for the customers.
- On the Cisco SD-WAN Manager **Settings** page, there is an option for **Symantec Automated** or **Cisco Automated** where automated refers to automatic submission of CSRs and retrieval of certificates. The option does include automation of certain steps of the process, compared to the manual option. However, the step to trigger the generation of CSRs for each controller is still manual, to be done by you, to initiate the renewal process.
- Note that the Cisco SD-WAN Manager Dashboard shows a warning 6 months in advance that the certificates are about to expire.
- You can view the expiry date at any time at by choosing **Configuration > Certificates > Controllers** from the Cisco SD-WAN Manager menu.



Note Starting from Cisco IOS XE Catalyst SD-WAN Release 17.13.1a, the **Controllers** tab is renamed as the **Control Components** tab to stay consistent with Cisco Catalyst SD-WAN rebranding.

- The Cisco CloudOps team sends email notifications 30/15/5 day prior to expiry, to the registered email address contact for the overlay in your system as well.

- You can open a case with us anytime to request the current registered email address or change it. We recommend that customers help keep the owner email address updated for all Cisco CloudOps notifications. We recommend keeping us updated with the customer contact email address for alert notifications, preferably a team mailer address instead of an individual user email address.
- Also, we recommend being aware of the controller certificate expiry dates and plan for renewal at least 1 month before expiry.



CHAPTER 4

Provisioning

- [Getting Access to Cloud Hosted Controllers, on page 25](#)
- [Cloud-hosted SD-WAN Control Components: Interfaces and access, on page 26](#)
- [Custom IP Prefixes for Cloud Hosted Controllers, on page 30](#)

Getting Access to Cloud Hosted Controllers

Cisco managed Cloud Hosted controllers are by default closed for management access. Cisco does not allow access to 0.0.0.0/0 to the Cloud Hosted Cisco Catalyst SD-WAN Controllers for security reasons. It is expected that you have specific public IP prefixes within your enterprise VPN that you access from and hence only those will be allowed to be opened for access. You can restrict access by requesting to allow only https and ssh to be on the allowed list, for your given source IP prefixes.

Cisco SD-WAN Cloud-Pro controllers have private IP addresses on their interfaces. Each private IP address has a 1:1 NAT mapped to a public IP address on the cloud. These IP addresses do not change irrespective of whether the interface is configured to use static IP or DHCP. The IP addresses only change when the instance is recovered or replaced.

The allowed-list is applied to all the network interfaces of all the controllers that have public IP addresses.

Update Inbound Rules

You can update the allowed-list applied to your Cisco SD-WAN Cloud-Pro controller set based on the overlay type.

1. **Shared tenant overlay:** To update or view the allowed-list applied to your Cisco SD-WAN Cloud-Pro controller set, open a case with Cisco TAC support.

You can request support for the following:

- Provide up to 5 IP prefixes to be allowed on the access-list
- Allow only https access to the IP prefixes for the web login to the Cisco SD-WAN Manager portal

2. **Dedicated Overlay:** To enable Cisco SD-WAN Cloud-Pro single tenant dedicated controllers to add, delete, or modify cloud security group allowed-lists, use one of the following options:

- You can login into the Cisco Catalyst SD-WAN Portal at <https://ssp.sdwan.cisco.com> and manage the access-list. You need to be the Cisco PNP Smart Account admin for the Smart Account where the overlay controller profile is based.

- You can provide up to 200 IP prefixes to be allowed on the access-list.
- You can open a Cisco TAC support case and provide the following information:
 - Overlay/VA name
 - Cisco SD-WAN Manager IP/FQDN
 - IP address
 - Specify whether to mark an IP address as allowed for all traffic or selected traffic (for example https, SSH, and so on).

Only the Smart Account administrator can access the Cisco Catalyst SD-WAN Portal which is used to view and perform operational tasks related to a customer's Cisco SD-WAN Cloud-Pro controller infrastructure, such as viewing the controllers' IP addresses and modifying the controllers' IP access lists. To disable SA administrator privileges for users, go to the Manage Smart Account section in [Cisco Software Central](#), and remove the users as Smart Account administrators. Alternatively, use the IDP (identity provider) onboarding feature to grant trusted users access to the Cisco Catalyst SD-WAN Portal.

Cloud-hosted SD-WAN Control Components: Interfaces and access

Cloud-hosted SD-WAN Control Components: Interfaces

Network interface allocation and configuration for SD-WAN Control Components

For SD-WAN Manager instances, Cisco allocates three network interfaces:

- eth0
- eth1
- eth2

For SD-WAN Validator and SD-WAN Controller instances, Cisco allocates two network interfaces:

- eth0
- eth1

The network interfaces are assigned with public and private IP addresses, which are static. However, they may change when an SD-WAN Control Component instance is replaced or moved to a new region.



Note You can view the static IP address using the [Catalyst SD-WAN Portal](#), from **Overlay Details > Controller view > Private IP**.

Configuring the interfaces

This table summarizes the recommended configurations for each interface.

Table 5: Interface configuration

Control Component	Network interface 1 (eth0)	Network interface 2 (eth1)	Network interface 3 (eth2)
	Management access	Control access by nodes in fabric	Communication among SD-WAN Manager instances in a cluster; SD-AVC component functionality
SD-WAN Manager	Private IP: Static, assigned by Cisco. (See note 1.) Public IP: Static, assigned by Cisco. 1-to-1 NAT mapping to private IP. Configuration requirements: • VPN 512 • Non-tunnel interface • Configure as a DHCP client (See note 2.)	Private IP: Static, assigned by Cisco. Public IP: Static, assigned by Cisco. 1-to-1 NAT mapping to private IP. Configuration requirements: • VPN 0 • Tunnel interface • Configure as a DHCP client (See note 2.)	Private IP: Static, assigned by Cisco. Public IP: None assigned. Configuration requirements: • VPN 0 • Non-tunnel interface • Configure with a static IP. Use private IP address that Cisco has assigned to the interface. View the static IP address using the Catalyst SD-WAN Portal , from Overlay Details > Controller view > Private IP .
SD-WAN Validator	Same as for SD-WAN Manager.	Same as for SD-WAN Manager.	Not applicable
SD-WAN Controller	Same as for SD-WAN Manager.	Same as for SD-WAN Manager.	Not applicable

Note 1: Accessible by a cloud gateway deployed using the Catalyst SD-WAN Portal. For details, see the section about [custom IP prefixes for cloud-hosted SD-WAN Control Components](#) in the *Cisco Catalyst SD-WAN CloudOps* guide.

Note 2: Even with DHCP configured for IP assignment, the interface always receives the same private IP on every DHCP renewal.

Cloud-hosted SD-WAN Control Components: Access

Edge device access to SD-WAN Control Components

WAN edge devices connect to SD-WAN Control Components only using the VPN 0 tunnel interface of the SD-WAN Control Components.

Edge device access to SD-WAN Control Components

Edge devices communicate with SD-WAN Control Components by transport layer security (TLS) or datagram transport layer security (DTLS) ports.

If you have an on-premises firewall, you can configure your firewall either

- to enable traffic on any IP (using 0.0.0.0) for these specific TLS or DTLS ports, or
- to enable traffic to the current public IP addresses of the cloud-based SD-WAN Controllers.



Note You can find the assigned public IP on the [Catalyst SD-WAN Portal](#), in **Overlay Details > Controller view > Public IP**.

For more information about TLS and DTLS ports, see [Ports Used by Cisco Catalyst SD-WAN Devices Running Multiple vCPUs](#) in the *Cisco Catalyst SD-WAN Getting Started Guide*.

Management access to SD-WAN Manager

Users connect to SD-WAN Manager, for management access, using fully qualified domain names (FQDNs) mapped to the VPN 512 public IP.

If the overlay is provisioned with a 3-node or 6-node SD-WAN Manager cluster, the FQDN resolves to the public IP addresses of all of the SD-WAN Manager instances.

HTTPS access for SD-WAN Manager

HTTP/HTTPS access is available only for SD-WAN Manager, not for other SD-WAN Control Components.

Domain names for SD-WAN Manager and SD-WAN Validator

In a Cloud-hosted SD-WAN environment, Cisco assigns a domain name only to SD-WAN Manager and SD-WAN Validator for cloud hosting.

Access SD-WAN Validator by domain name

When configuring nodes in the SD-WAN fabric, do not configure access to SD-WAN Validator by IP address. Use the FQDN of the SD-WAN Validator instead. This will ensure continued reliable operation in case the SD-WAN Validator IP addresses change or more SD-WAN Validators are added to the fabric.

DNS server

We recommend configuring a DNS server accessible in VPN 0 for each node in the fabric, including hardware edge devices, software edge devices, and the SD-WAN Control Components.

Example:

```
vpn 0
  dns 208.67.222.222 primary
  dns 208.67.220.220 secondary
```

Examples of correct and incorrect configurations for access to SD-WAN Validator

These examples show how to configure access to SD-WAN Validator on other nodes in the network, such as SD-WAN Manager, SD-WAN Controllers, and edge devices.

Correct

This configuration is correct because it uses the SD-WAN Validator domain name:

```
system
  vbond validator-domain-name
```

Incorrect

This configuration is incorrect because it uses a static IP address rather than a domain name:

```
system
  vbond 10.1.1.1
```



Note

You can view the FQDN of the SD-WAN Validator for your fabric using the Catalyst SD-WAN Portal. Open **Overlay Details > Description > vBond DNS**. Note that **vBond** may be replaced by **SD-WAN Validator**.

Example of correct and incorrect configurations of VPN 0 for access to SD-WAN Validator

These examples show the VPN 0 configuration to enable access to SD-WAN Validator on other nodes in the network, such as SD-WAN Manager, SD-WAN Controllers, and edge devices.

Correct

This configuration is correct because it configures a DNS server that can resolve the SD-WAN Validator domain name.

```
vpn 0
  dns dns-server-ip primary
```

Incorrect

This configuration is incorrect because it uses a static host IP for SD-WAN Validator:

```
vpn 0
  ip host validator-domain-name 10.1.1.1
```



Note You can view the FQDN of the SD-WAN Validator for your fabric using the Catalyst SD-WAN Portal. Open **Overlay Details > Description > vBond DNS**. Note that **vBond** may be replaced by **SD-WAN Validator**.

Custom IP Prefixes for Cloud Hosted Controllers



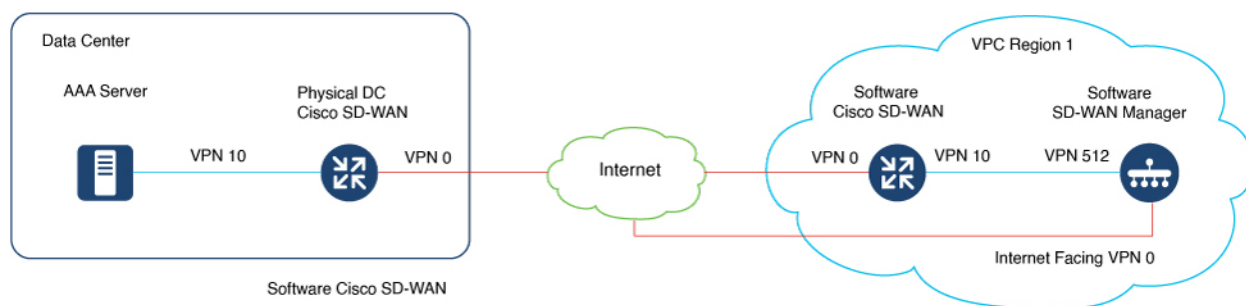
Note Custom IP prefixes are applicable only if you use Cisco-hosted, cloud-based, dedicated single tenant controllers. These are not applicable for shared tenant overlays.

There are certain use-cases, where you may need custom network prefix-based IPs on the cloud controller interfaces for management access and control. For example,

- To access the management VPN 512 of Cisco SD-WAN Manager and Cisco SD-WAN Validator or Cisco SD-WAN Controller devices over Cisco Catalyst SD-WAN tunnel with AAA or TACACS based authentication.
- To send syslog from Cisco SD-WAN Manager over VPN 512 to a syslog server over Cisco Catalyst SD-WAN tunnel.

Figure 2: AAA TACAS

User1, User2 --- AAA --- [vpn10] VE(DC) [vpn0] --- [vpn0] INTERNET [vpn0] --- [vpn0] Edge(cloud) (each region) [vpn10] --- [vpn512] SD-WAN Manager [vpn0]



By default, the Cisco-managed cloud hosted controllers are deployed with 10.0.0.0/16 based subnets, including the VPN 512 subnet. If you add the cloud Cisco Catalyst SD-WAN and bring the VPN 512 subnet as a reachable subnet within your fabric, it might conflict with an existing subnet.

In such cases, you need to share a /24 prefix for each of the two regions of deployment of controllers. These IP prefixes are used to create the controllers, and the subnets are then configured to be available within the Cisco Catalyst SD-WAN fabric.

Request for Cloud Gateways for Post Overlay Provisioning:

Open a case for CloudOps at TAC-CSOne with the following details:

1. To enable AAA or TACACs, you need to provide IP prefixes, unused within your existing fabric, that you can use to create the controllers (original controllers are shutdown, snapshotted, and cloned back).
Each region in which the controllers are set up has one /24 Cisco Catalyst SD-WAN fabric wise unique custom subnet. Each overlay has two regions, so we need two subnets.
2. Admin credentials to the Cisco SD-WAN Validator, Cisco SD-WAN Controller and Cisco SD-WAN Manager devices.
You can provide credentials at the start of the actual change window.
3. You can schedule eight-hour maintenance window after the preapproval and prechecks completed by CloudOps engineer.
4. Enable DNS for Cisco SD-WAN Validator and configure all the controllers prior to start of the process.
5. Ensure that GR is set to default of 12 hours or more on Cisco Catalyst SD-WAN or Cisco SD-WAN Controller devices.
6. Reserve two available Cloud Cisco Catalyst SD-WAN UUIDs through PNP and attach to Cisco SD-WAN Manager.
7. Supported only for Single Tenant Single Node Cisco SD-WAN Manager overlays and Single Tenant Cluster Node Cisco SD-WAN Manager overlays for provisioned controllers, and all new to-be-provisioned controller sets. This feature is not supported for Cisco Multi-tenant Cisco SD-WAN Manager cluster overlay.
8. It is recommended that Cisco SD-WAN Manager have templates attached to Cisco SD-WAN Validator, Cisco SD-WAN Controller, and if existing cloud Cisco Catalyst SD-WAN devices from Cisco.

Configuration of Cloud Gateways Post Cisco Provisioning

1. Once Cisco CloudOps has completed the provisioning of the cloud gateways next to the cloud hosted controllers, CloudOps shares the public & private IP assignments for each cloud gateway to the customer. They are in the format (VPN 512, VPN 0, VPN X).
Cisco CloudOps will share the credentials for the newly provisioned cloud gateways.
2. The cloud gateways have their VPN 512 & VPN X interfaces in the same subnet as the VPN 512 of the controllers in that region.
The cloud gateways provisioned by the Cisco CloudOps are specifically for the AAA/TACACS purpose and always created in the above network layout format.
If there are any reachability issues to the cloud gateway, the issue generally lies with the interface IP or route configurations in the cloud gateway.
3. Also, note that the public & private IPs are 1:1 NAT'd and assigned to the cloud gateway interfaces. The gateway interface itself may be configured with dhcp, but it will always get the same IP from Cloud.
For VPN X interfaces, you will need to configure the static IP, exactly as the one shared by Cisco CloudOps.
Random IPs within the subnet cannot be used.
4. The cloud gateways are subject to the same Inbound allowed access-list as the controllers, as they are provisioned in the same unique environment per overlay.
You must login via SSH to the gateway public IPs and the credentials provided.

5. You must now configure the new cloud gateways with the necessary configurations. For example, site-id, system IP, organization name, Cisco SD-WAN Validator DNS or IP, and so on.
6. If you are using Enterprise root-ca, then you must upload and install the same on the cloud gateways as well.
7. You may configure AAA/TACACS on the Cisco SD-WAN Manager with auth-fallback to local with local having the vptelatac/ciscotacro/ciscotacrw user enabled. This allows Cisco support to login and troubleshoot issues when required.
8. You would need to acquire an unused cloud gateway UUID from the device list of the Cisco SD-WAN Manager, one per cloud gateway provisioned.

If you don't have any cloud gateway UUID available in the WAN Edge Device list on your Cisco SD-WAN Manager, then you may need to login into the Cisco PNP portal, on the overlay's associated Smart Account and Virtual Account, and Add Software Devices (VEDGE-CLOUD-DNA) and then Sync Smart Account on the Cisco SD-WAN Manager.

9. You must then activate the UUID on the cloud gateways to allow them to be authenticated by the Cisco SD-WAN Manager and join the Cisco Catalyst SD-WAN fabric.
10. You must configure the controllers' (Cisco SD-WAN Manager, Cisco SD-WAN Validator, Cisco SD-WAN Controller) VPN 512 with a specific static route for customer's Enterprise subnets (from customers admin team intend to access the controllers for management) to point to the cloud gateway's VPN X static IP.
11. For an overlay hosted by Cisco on Azure, please open a Cisco TAC case and provide the specific enterprise subnet prefixes, from where the connectivity to the VPN 512 of the controllers is required.

The Azure subnet default gateway is the defacto gateway even if you configure the gateway service VPN IP to be the gateway for your enterprise subnets. Hence in addition to your configuration on VPN 512 on the controllers, there is additional configuration needed on the Azure side. Cisco will help apply an Azure Route Table (RT) entry for each of the necessary Enterprise subnets and also enable IP forwarding on the cloud gateway interfaces.



CHAPTER 5

Monitoring



Note

To achieve simplification and consistency, the Cisco SD-WAN solution has been rebranded as Cisco Catalyst SD-WAN. In addition, from Cisco IOS XE SD-WAN Release 17.12.1a and Cisco Catalyst SD-WAN Release 20.12.1, the following component changes are applicable: **Cisco vManage** to **Cisco Catalyst SD-WAN Manager**, **Cisco vAnalytics** to **Cisco Catalyst SD-WAN Analytics**, **Cisco vBond** to **Cisco Catalyst SD-WAN Validator**, **Cisco vSmart** to **Cisco Catalyst SD-WAN Controller**, and **Cisco Controllers** to **Cisco Catalyst SD-WAN Control Components**. See the latest Release Notes for a comprehensive list of all the component brand name changes. While we transition to the new names, some inconsistencies might be present in the documentation set because of a phased approach to the user interface updates of the software product.

- [Monitor the Cisco Catalyst SD-WAN Cloud-Hosted Controllers, on page 33](#)

Monitor the Cisco Catalyst SD-WAN Cloud-Hosted Controllers

Cloud hosted controller monitoring covers the following:

- Infrastructure monitoring of the following:
 - CPU and data disk utilization.
 - Loss of connectivity to network interfaces.
 - Failure to reach instances.
- Service monitoring of the following:
 - Expiration of controller SSL certificates.
 - Availability of the Cisco SD-WAN Manager web server.
 - Loss of control connection to the controllers.

Health Monitoring of Overlays with Cisco SD-WAN Manager Version below 20.3.x

The cloud monitoring is performed as a part of the Cisco Catalyst SD-WAN cloud-hosting services to ensure the availability of the Cisco SD-WAN Controllers. By default, Cisco SD-WAN Manager is configured with a user called `viptelatac` with `operator` privileges. Cisco uses this user to login to Cisco SD-WAN Manager and to collect and monitor the health of Cisco Catalyst SD-WAN.

The Cisco SD-WAN Manager audit log displays periodic logins from the monitoring system using the `viptelatac` user. The monitoring service uses RestAPIs to collect health information from Cisco SD-WAN Manager.

In case you want to disable the Cisco cloud monitoring system, you can open a Cisco TAC case with the Cisco Catalyst SD-WAN Cloud Infra team, requesting to disable the cloud monitoring. Once the monitoring is disabled, you can also remove the configured `viptelatac` user from the Cisco SD-WAN Manager.

Cisco Cloud Infra team will also use the `viptelatac` user to login to the Cisco SD-WAN Manager to do additional health checks, triage issues in response to internally generated alerts, as well as to assist with customer opened TAC cases.

Health Monitoring of Overlays with Cisco SD-WAN Manager Version Running at or above 20.3.x

From Cisco SD-WAN Release 20.3.1 release onwards, push based model is used.

In this model, the monitoring architecture uses Cisco SD-WAN Manager to authenticate with the system to send the health data. Cisco SD-WAN Manager pushes the data instead of monitoring system logging into the Cisco SD-WAN Manager with the `viptelatac` user. In order for this to work, you need to explicitly provide consent on the Cisco SD-WAN Manager settings page, as well as configure a One Time Password (OTP). The `viptelatac` user is not needed once Cisco SD-WAN Manager is upgraded to 20.3.1 or above.

You can login to Cisco SD-WAN Manager and perform the following steps:

1. Go to **Settings > Cloud Services > Enable**
2. Enter the OTP value. You can request the token from the Cisco CloudOps team by opening a Cisco TAC Support case.
3. Leave the Cloud Gateway URL blank.
4. Check the **vMonitoring** to enable monitoring.
5. Approve permission to collect the data regarding health status of the overlay from Cisco SD-WAN Manager.

For version 20.3.x and above, Cisco Cloud Infra team will use the `ciscotacro` and `ciscotacrw` user to login to the Cisco SD-WAN Manager to do additional health checks, triage issues in response to internally generated alerts, as well as to assist with the customer opened TAC cases. The same user will also be used to perform automated infrastructure upgrades and certain software updates based on pre-notified changes to the customer contacts for the overlay.

The `ciscotacro` user has read-only `operator` group privilege while `ciscotacrw` has read-write `netadmin` group privilege. For certain enhanced debugging, cloud infrastructure upgrades and management, Cisco Cloud Infra team needs to use the `ciscotacrw` user.

Only specific Cisco support teams have the ability to login via these users and they are based on a token challenge and token response based password mechanism i.e., the two users are not based on static passwords.

In case, you want to disable this access on any of the Cisco Catalyst SD-WAN fabric controllers, you can remove the user from the configuration at any time. However, this will limit Cisco ability to triage the issues.

Alert Notifications by CloudOps

CloudOps team manages the infrastructure of the cloud hosted instances and help with the monitoring and backend infrastructure maintenance. However, CloudOps team does not make changes or manage the running software version or configuration of the instances.

CloudOps team may send alert notifications to customers, based on any issues seen, which may indicate either software issue or misconfiguration or some features overutilizing the capacity, which CloudOps team is not aware of. Customers may be running their own tests, changes, or configuration updates, that the team is not aware of.

CloudOps team will therefore, only notify the customers instead of taking direct action on the hosted controller instances, and request customer to open a Cisco TAC support case for assistance and evaluation as needed. Once the customer has a TAC case open, Cisco TAC and thereafter, CloudOps team, can work together with the customer, to resolve the issue as needed.

Update Overlay Contact for Receiving Alert Notifications

- Every Cisco provisioned cloud-hosted overlay has a single customer contact email address registered as the owner, to receive CloudOps Alert notifications.
- By default, the contact email address provided on the Cisco Sales Order's End Customer details has been used as the owner contact.
- Customers can open a Cisco TAC case to review or update the contact at any time.
- For Cisco-hosted, cloud-based, dedicated, single tenant controllers, you can directly update the owner contact email address through the [Cisco Catalyst SD-WAN Portal](#).
- We support only one email address contact as the owner contact and hence it is recommended that you provide a group mailing list email address.



CHAPTER 6

Cloud Infrastructure



Note

To achieve simplification and consistency, the Cisco SD-WAN solution has been rebranded as Cisco Catalyst SD-WAN. In addition, from Cisco IOS XE SD-WAN Release 17.12.1a and Cisco Catalyst SD-WAN Release 20.12.1, the following component changes are applicable: **Cisco vManage to Cisco Catalyst SD-WAN Manager**, **Cisco vAnalytics to Cisco Catalyst SD-WAN Analytics**, **Cisco vBond to Cisco Catalyst SD-WAN Validator**, **Cisco vSmart to Cisco Catalyst SD-WAN Controller**, and **Cisco Controllers to Cisco Catalyst SD-WAN Control Components**. See the latest Release Notes for a comprehensive list of all the component brand name changes. While we transition to the new names, some inconsistencies might be present in the documentation set because of a phased approach to the user interface updates of the software product.

- [Cisco Cloud-Hosted Controller Snapshots, on page 37](#)
- [Cisco Catalyst SD-WAN Analytics, on page 38](#)
- [Pen Test, on page 38](#)
- [Mandatory Maintenance of Cloud-Hosted Controllers, on page 38](#)
- [Cisco Catalyst SD-WAN Disaster Recovery Guidelines, on page 39](#)

Cisco Cloud-Hosted Controller Snapshots

Cisco takes regular snapshots of the cloud hosted Cisco SD-WAN Manager controller managed by Cisco, based on the snapshot frequency. The snapshot frequency is set by default to once every day, typically at midnight of the region of deployment, and the last 7 snapshots are retained. The snapshot frequency can be configured from once a day up to once in 4 days. For more information on Snapshots, see [Information About Snapshots](#).

You can open a Cisco TAC support case with the Cisco CloudOps team to review the current snapshot setting or change it on the Cisco Catalyst SD-WAN Portal. You can retain only a maximum of last 7 periodic snapshots. The Cisco Catalyst SD-WAN Controller and Cisco SD-WAN Validator are stateless, so snapshots are not taken. We recommend that their configurations be done via templates on Cisco SD-WAN Manager for disaster recovery.

You cannot download the snapshots, as snapshots are stored within the Cisco cloud account. However, you can download the config-db backup file from Cisco SD-WAN Manager and save the configurations including templates using command [request nms configuration-db backup path](#).



Note Since Cisco SD-WAN Validator and Cisco Catalyst SD-WAN Controller are stateless, snapshots are not captured. Use Cisco SD-WAN Manager template to configure and save Cisco SD-WAN Validator and Cisco Catalyst SD-WAN Controller Configuration settings.

Take an On-demand Snapshot



Note The on-demand snapshot process is applicable only for overlays with Cisco-hosted, cloud-based, dedicated, single tenant controllers. This is not applicable if you have a shared tenant overlay.

For any major planned change windows for Cisco SD-WAN Manager, You can take on-demand snapshot using Cisco Catalyst SD-WAN Portal. This can be requested via opening a Cisco TAC support case with the Cisco CloudOps team. You need to freeze the configuration changes and allocate up to eight hours prior to the change window to allow the on-demand snapshot to be taken and completed. We can store up to one on-demand snapshot. We can store this on-demand snapshot for a period of 10 days from the date of creation of the snapshot. Also, each time a new on-demand snapshot is taken, the previous one, if present, is automatically removed and replaced with the new one.

Cisco Catalyst SD-WAN Analytics

Refer to [Cisco Catalyst SD-WAN Analytics](#).

Pen Test

Customers with overlay controllers in AWS can conduct their own pen tests for the Cisco Catalyst SD-WAN solution without approval using:

- <https://aws.amazon.com/security/penetration-testing/>

Customers with overlay controllers in Azure can conduct their own pen tests for the Cisco Catalyst SD-WAN solution without approval using:

- <https://www.microsoft.com/en-us/msrc/pentest-rules-of-engagement>

Mandatory Maintenance of Cloud-Hosted Controllers

The Cisco CloudOps team sends email notices to customers for mandatory reboot of a specific cloud-hosted controller managed by Cisco, only when hosted in AWS. Sometimes, maintenance is required on the instances and rebooting the instances prior to the cloud provider's maintenance window. It allows you to move them from the current hardware node requiring maintenance, to a new healthy hardware node, to avoid disruption of service.

The Cisco CloudOps team then sends notifications to the registered email address of the customer, which is a single email address registered for an overlay within the Cisco CloudOps system. Note that this registered

email address is configured initially using the original Sales Order's **End Customer Email Address** field and can be updated anytime by logging into the Cisco Catalyst SD-WAN Portal (link to <https://ssp.sdwan.cisco.com>). This registered email address is not derived from Cisco SD-WAN Manager Settings page.

You can reschedule to update the change window, as long as the requested date and time is before the cloud provider's maintenance window time. The amount of advance notice is not guaranteed and depends on the severity of the issue on the hardware node on cloud provider side.

Cisco Catalyst SD-WAN Disaster Recovery Guidelines

- Cisco Catalyst SD-WAN disaster recovery is based on Cisco SD-WAN Manager disk volume snapshots or configuration database backups.
- These configuration database backups and volume snapshots are taken each daily, typically around midnight time of the location of the Cisco SD-WAN Manager instance and securely stored on cloud.
- For Cisco SD-WAN Release 20.3.x and later, you can turn off configuration database backup feature, if desired, and take own backups and make them available to CloudOps when needed for recovery of the service.
- Cisco SD-WAN Manager disk volume snapshots are taken every night and sometimes on-demand for customer request or at start of major change windows. Each Cisco SD-WAN Manager has two or more disks and a snapshot of each of the volumes is taken at the exact same time to form an overall backup of the Cisco SD-WAN Manager instance. The snapshots, once completed, in the region where the Cisco SD-WAN Manager is running, are then copied over to the designated backup region, usually a different geographic region.

For example, Cisco SD-WAN Manager may be running in US-East and backup region may be designated as US-West. The backup region is essentially the same region, where the second Cisco Catalyst SD-WAN Validator and Cisco Catalyst SD-WAN Controller are already running.

- Cisco Catalyst SD-WAN Validator and Cisco Catalyst SD-WAN Controller are stateless services and have a CLI-managed configuration or Cisco SD-WAN Manager provide configuration and hence they aren't backed up.
- There's no standby or active Cisco SD-WAN Manager service in a backup region. Three or six node cluster offers high availability of Cisco SD-WAN Manager, running within the same availability zone and region.
- Cisco Catalyst SD-WAN Validator and Cisco Catalyst SD-WAN Controller services are deployed in primary and backup regions. Both Cisco Catalyst SD-WAN Validator and Cisco Catalyst SD-WAN Controller work in active mode. The device and policy information is pushed to both the instances from Cisco SD-WAN Manager. When one region fails, Cisco Catalyst SD-WAN Controller and Cisco Catalyst SD-WAN Validator continue to function fine in the backup region.
- Cisco Catalyst SD-WAN is designed for the data plane to continue to function even if all the controllers fail. GR (Graceful Restart) timer configuration enables the high availability of the data plane. GR timer is configured to hold on to the routes advertised by Cisco Catalyst SD-WAN Controllers, by default for 12hrs. We recommend Cisco Catalyst SD-WAN customers to choose GR timer value judiciously to allow controllers to be back up in case of failures, and at the same time be able to learn the new routes from changed network configurations.
- Configuration database-based recovery method when used, allows only templates and policies to be restored. Volume-based recovery is used to include the stats data collected as well.

Volume Snapshot based Recovery Process

- Once determined that the Cisco SD-WAN Manager instance needs to be replaced with a backup, we can initiate the Disaster Recovery (DR) process.
- For DR at same region, Cisco pick the same region and same datacenter as the existing Cisco SD-WAN Manager instance location.

We also specify the time/date of the snapshot set to use, based on requirement and availability.

- Once DR triggers, the system first shuts down the existing Cisco SD-WAN Manager instance.
- System then uses the volume snapshots to create a new cloud instance with the same set of disks, same instance size specifications, same private subnets, same security access-list, same isolated environment as that of the original Cisco SD-WAN Manager. Once the instance is up, the system swaps the public IPs from the old shutdown Cisco SD-WAN Manager instance to the new Cisco SD-WAN Manager instance.
- Overall, the new running Cisco SD-WAN Manager instance has the same public IPs, but new private IPs and have the same software version, same configuration, same data, as that present at the time when snapshot is taken.
- Cisco SD-WAN Manager has the necessary information to join the fabric. You can use the same FQDN/URL to log in into the Cisco SD-WAN Manager instance as before.
- For DR to the backup region, in the unlikely case where the primary region of Cisco SD-WAN Manager has failed and unavailable, we use the exact same process, except that the backup cloud region is selected.
- The difference with DR to backup region is that, once the new Cisco SD-WAN Manager instance is running in backup region, there's no swapping of public IPs from old region to new region. Cloud regions have a specific public IP pool per region and can't be assigned to instances across regions.

Therefore, the new DR Cisco SD-WAN Manager instance in backup region, has new public IPs. The system updates the FQDN/DNS with the new public IP of the Cisco SD-WAN Manager.

In this case, you may need to update the enterprise end firewall with the new public IP of the Cisco SD-WAN Manager.

Configuration Database backup by Cisco CloudInfra System

- Prior to Cisco vManage Release 20.3.1, the configuration database was backed up only if:
 - Monitoring is enabled in Cisco CloudInfra system. If 'viptelatac' user is unusable on the Cisco SD-WAN Manager for any reason, monitoring gets disabled, and customers are notified with request for correction.
 - The 'viptelatac' user must be usable on the Cisco SD-WAN Manager.
 - The configuration database size is lesser than 4GB.
- Cisco vManage Release 20.3.1 and later, the configuration database is backed up only if:
 - Monitoring is enabled in Cisco CloudInfra system.

**Note**

In Cisco SD-WAN Manager, if the cloud service is disabled for any reasons, then monitoring gets disabled on Cisco CloudInfra system and customers are notified with request for correction.

- From the Cisco SD-WAN Manager menu, choose **Administration > Settings**, and enable Cloud Services and vMonitoring along with OTP added in the same section.
- In the Cisco SD-WAN Manager CLI, the **nms configuration-db daily-backup** service is enabled.
- The configuration database size is lesser than 4GB.

Configuration Database based Recovery Process

- If volume snapshot is not viable for DR for any reasons, then Cisco uses the configuration database recovery process. Cisco creates a brand new Cisco SD-WAN Manager instance and use the configuration database backup to restore the original configuration files. With this method, the statistics database of the original Cisco SD-WAN Manager instance is not restored. This method restores your templates and policies configuration. The new Cisco SD-WAN Manager instance in this case has both new public IPs and new private IPs.
- We update the FQDN/DNS of the Cisco SD-WAN Manager to use the new public IP of the new instance.
- In this case, you may need to update the enterprise end firewall with the new public IP of the Cisco SD-WAN Manager.
- The process for using disaster recovery method using configuration database backup remains same for both same region and backup region recovery.
- For process details, see [Troubleshooting TechNotes](#).



APPENDIX **A**

Opening a Cisco TAC Support Case for the Cisco Catalyst SD-WAN Cloud Infra team

Procedure

- Step 1** Go to <https://mycase.cloudapps.cisco.com/case>
- Step 2** Select **New Case** > **Products & Services** > **Open Case**.
- Step 3** Enter the appropriate entitlement information. You can use the serial number of a WAN Edge device for entitlement information.
- Step 4** Select **Next** and enter your case details.
- Step 5** Select **Technology** > **Search** and enter the appropriate Tech and Sub Tech keywords:
- Technology: SDWAN - Cisco-Hosted
 - SubTechnology: SDWAN Cloud Infra
-

